A Navigation and Mapping Method for UAS during Under-the-Canopy Forest Operations

Adam Schultz, Russell Gilabert, Akshay Bharadwaj and Maarten Uijt de Haag
Ohio University, Athens, OH, USA

Zhen Zhu
East Carolina University, Greenville, NC

***Abstract*— This paper discusses the preliminary design and implementation of a simultaneous navigation, and mapping method for operation of multi-copter unmanned aircraft systems (UAS) in forest environments under the canopy. A data collection platform has been developed using a 3DR hexacopter platform equipped with laser scanners, cameras, GNSS and inertial sensors to observe the environment and used as an input to the SNAM methods. Details on this flight prototype are addressed. Furthermore, the paper will include preliminary flight test results of our hexacopter UAS operating in a forest environment. In addition to these discussions, the paper will address some application-specific issues, specifically with respect to biological, civil engineering and archeological applications.**

Keywords—sensor integration; Global Positioning System; inertial navigation; electro-optical sensors; laser range scanners; iterative closest point; simultaneous localization and mapping.

I. INTRODUCTION

The number of applications for Unmanned Aerial Vehicles (UAVs) or UAS (Unmanned Aircraft Systems) is increasing every day from higher altitude operations such as surveillance to low altitude operations such as agriculture, mapping, deliveries, search and rescue and law enforcement. Among the low altitude operations, UAS may have to operate in a challenging environment in terms of obstacles (e.g. urban, forest) or in terms of navigation. This paper focuses on low altitude UAS forest operations under the canopy (foliage) to support various types of applications, such as search for sinkholes or forgotten mines, drainage, and mapping; forestry and biological applications; or even archeological applications. In these under-the-canopy forest operations, the use of the Global Navigations Satellite Systems (GNSS) often results in unreliably position and velocity solutions, which, even if available, have a lower accuracy. In addition, an absolute global position solution such as GNSS does not take into account the typical environment in which the UAS operates and can collide with. An alternative solution is introduced in this paper, which combines GNSS with a simultaneous navigation, and mapping (SNAM) capability. SNAM is used instead of Simultaneous Localization, and Mapping (SLAM), as prediction of the platform with respect to the environment plays a very important role in under the canopy operations.

A SNAM algorithm has been developed that makes use of multiple onboard laser scanners. Since UAS operations are the focus, 6 degrees of freedom (or pose) must be estimated from available observed environmental features that change as a function of platform altitude. Improvement of algorithm observability by intentional motion patterns such as height changes and roll and pitch motion will be addressed. The proposed SNAM algorithm is, furthermore, aided by low-cost inertial measurement unit that helps in the feature association process and the filter's prediction phase. Where available, GNSS measurements will be screened for quality and included in the solution, so the obtained map of the environment can be geo-referenced in a global frame. As the map is built online, conflict-free trajectories must be established in addition to the intentional motion, to avoid the trees while mapping the forestry environment or performing other missions at hand.



Figure 1. Multi-copter operating in the target environment.

A data collection platform has been developed using a 3DR multi-copter platform. The platform has been equipped with laser scanners, cameras, GNSS and inertial sensors to observe the environment and as an input to the SNAM methods. All data collection software as well as parts of the algorithms have been implemented in the Robotic Operating Systems (ROS) on an onboard embedded processor

To enable operation of UAVs at any time in any environment, a precision navigation, attitude, and time capability is required that is robust and not solely dependent on GPS. In aerial robotics, an estimate of a platform's 6 degrees-of-freedom (6DOF) is required: 3D position (x,y,z), attitude (φ,θ) and heading (ψ). To improve availability and guarantee continuity of service in GPS-challenged environments, GPS can be integrated with an IMU [1] or improved by increasing its sensitivity by using external data sources (i.e. assisted GPS). To achieve this in the absence of GPS, many alternatives to GPS may be considered and have been investigated. Some of which include (i) the integration of laser scanners and/or imagery with an inertial sensor [1], (ii) the use of signals of opportunity [2],

978-1-5090-2042-3/16/$31.00 © 2016 IEEE

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.

(iii) and beacon-based navigation (i.e. pseudolites, UWB, etc.) [3]. The integration strategy addressed in this paper falls under category (i) and combines measurements from GNSS, an IMU, and multiple 2D laser scanners. GNSS provides measurements suited for estimation of the position and the IMU provide measurements that can be used to estimate all 6 unknowns.

Two-dimensional (2D) laser scanners have been used extensively in robotics to estimate 3 degrees-of-freedom (3DOF), (x, y, ψ) for 2D navigation; for example, [5] and [6] describe methods that monitor features such as lines or points to estimate a platforms 2D position and heading or aid an inertial navigator. Alternatively, many research papers have addressed laser scanner-based 3DOF SLAM methods such as the grid-based SLAM method described in [7] or methods that use some form of scan-matching such as the Iterative Closest Point (ICP) approach [8]. Specifically, [9] describes a method for ground-based SLAM in an outdoor park environment using a laser range scanner. In their approach the laser scanner observations for each tree are converted to a single point feature and then input to an extended Kalman filter (EKF) that has the feature locations (i.e. map) modelled as part of the filter's state vector.

Extension to the use of 2D laser scanners for 3D pose of, for example, aerial vehicles, has been proposed in numerous papers. The method described in [10] uses the aerial platform itself to rotate the 2D laser scanner and, assuming a structured environment, uses planar feature to estimate the 3D pose estimation of the platform. SLAM methods have also been developed for 6DOF platform operation such as the methods described in [11][12][13] and [14]. For example, in [14] an estimate of the horizontal position (x, y) and the platform heading (ψ) is obtained from a combination of laser scan matching and grid-based SLAM using inputs from an IMU to level the laser measurements; the height (h) is estimated from a downward pointing laser scanner, and the pitch and roll (ϕ, θ) are obtained directly from the IMU. Parts of this method are used in proposed SNAM capability as explained in the next sections.

Use of UAV laser range measurements (i.e. LIDAR) for under the canopy mapping purposes in a post-processing manner has been addressed in [15]. Using the laser measurements for actual 3D UAV localization in a forest environment has been discussed in [16]. In the latter paper the authors propose a method that consists of a front-end and back-end module. In the front-end module incremental 2D pose estimates are obtained using laser scan matching and integrated with IMU acceleration and barometric height measurements in a Kalman filter to obtain the 3D position estimate. In the back-end module graph-based optimization methods are used to remove the long term drift and obtain an accurate map. This method show many similarities to the method used in the paper. Other papers have described methods to navigate through a forest environment using visual measurement form onboard cameras [17] and [18].

The method proposed in this paper builds on concepts discussed in [12][14] and [16] but extends the operational scenario to include the outside environment where GNSS is available allowing the inclusion of GNSS measurements. Furthermore, two additional low-range 2D laser scanners have been included. One scanner is pointing downward and scans in the platforms cross-track direction (yz-plane). The second has a scan plane that scans at in a slanted direction and lies between the scans of the downward and forward scanner laser scanners. This installation orientation is illustrated in Figure 2 and aims to increase the instantaneous field of view and get a good observation of the terrain underneath the canopy.

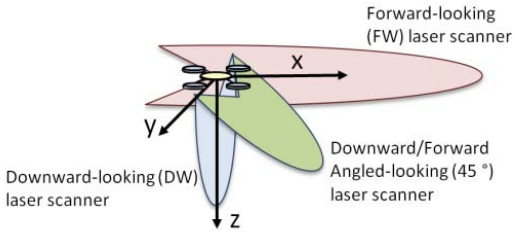


Figure 2. Installation orientations of the two 3D laser scanners

The next section describes some of the characteristics of the operational environment and the resulting constraints on the navigator design. Section 3, 4 and 5 will then then describe the proposed method. Finally, the platform and preliminary results are discussed in Sections 6 and 7.

II. NAVIGATING IN FORESTS

The large variation in types of forest is illustrated in Figure 3. Pine forests shown in the top-left quadrant are typically more structured as the tree does not change its shape in the z-axis. Furthermore, the figure shows an example of a flat area with tree distributed evenly across the terrain.



Figure 3.Various types of forest: Pine (top-left), temperate (top-right), tropical (bottom-left), hilly (bottom-right).

Temperate forests, depicted in the top-right quadrant) are a less structured as the trees tend to vary along the z-axis. This will

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.

have an effect on the feature location within the map as illustrated by Figure 4.

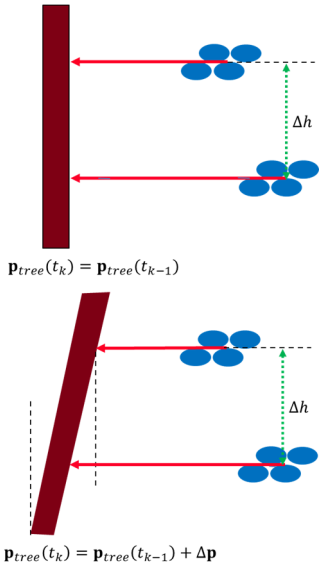


Figure 4. Effect of a "leaning" straight tree on the feature location within the map.

The area is typically relatively flat but has a higher obstacle density consisting of smaller branches, brush and other foliage that may pose a challenge to finding a conflict free trajectory. Tropical forests are highly non-structured as the foliage is very dense and obstacle avoidance may prove to be challenging as shown in the lower-left quadrant.

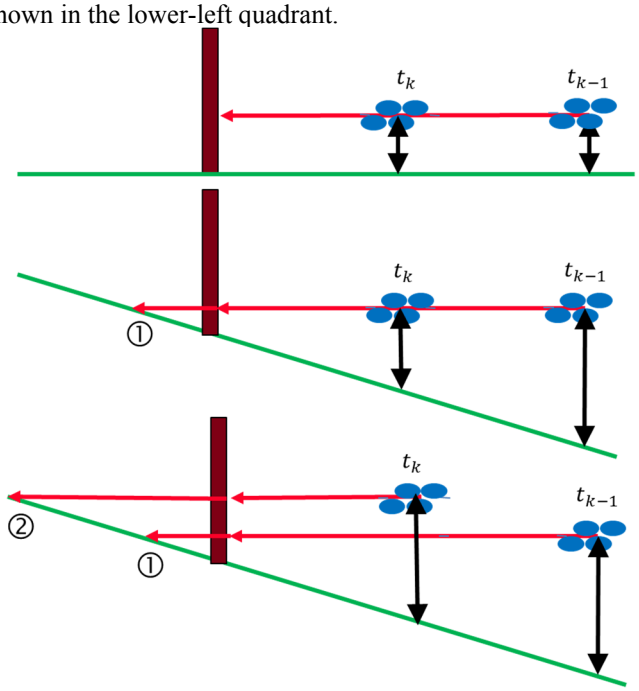


Figure 5. Effect of the underlying terrain on the sensor observations.

Finally, the terrain underneath the canopy may have different levels of relief (i.e. the forest maybe located on a hill). In that case the hill itself may be observable by the laser scanner as the intersection of the laser scan with the hill as depicted by ① in Figure 5. The height above ground level also change not due to a change in platform height, but due to the terrain itself, making it hard to distinguish between a change in platform height and a change in terrain height without the use of a baro-altimeter.

III. METHODOLOGY

The high-level functional block diagram of our approach is depicted in Figure 6. The proposed method for guiding, navigating and controlling the unmanned multi-copter consists of two main functions: the **tactical** Guidance Navigation and Control (GNC) function and the **strategic** GNC function [19].

The tactical GNC function makes optimal short-term trajectory planning decisions at a high update rate to accommodate a possible high speed of operation of the aerial vehicle in a potentially cluttered environment. These decisions are based on vehicle state (e.g. position, velocity, and attitude) updates from the navigation module and a local map of the environment. This map describes attributes of the local environment as perceived by the sensors that can be used in the path planning and collision avoidance process. Example maps in the context of this paper are forest maps and time-to-contact maps.

The strategic GNC keeps track of the planned and actual flight trajectories and provides the pilot with the next waypoint(s) that must be reached. In unknown environments, the strategic GNC function keeps track of a rough estimate of the platform trajectory and associated global map of the environment. Based on this information and the goals defined by the specific mission (i.e. outdoor/indoor navigation and mapping mission), this function will provide the tactical GNC algorithms with constraints regarding its path planning decisions (i.e. preferred direction of travel, decision between multiple similar choices such as doors or windows) and associated control behaviors. The global map can be formed using various typical SLAM back-end methods like composition of the global map from multiple local maps using for example Hierarchical SLAM [20] or GraphSLAM methods (e.g. [21]).

A block diagram of the tactical GNC navigator and mapping method is shown in Figure 7. When GNSS satellites are available, the navigator is based on tightly coupled GNSS/INS based on the algorithm proposed in [1]. In addition, this integration structure will include a baro-altimeter input. This addition will allow for calibration of the altimeter in the presence of GPS. The GPS/INS estimator uses a separate estimator for 3D dynamics (c) and position (d), the former is solely driven by carrier-phase measurements.

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.

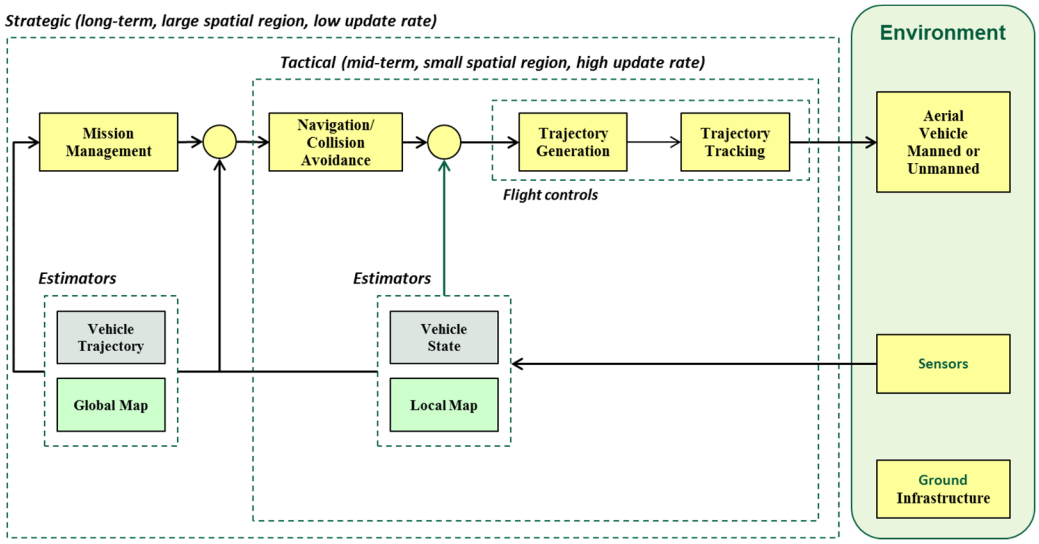


Figure 6. High-level functional block diagram of the hierarchical guidance, navigation and control function.

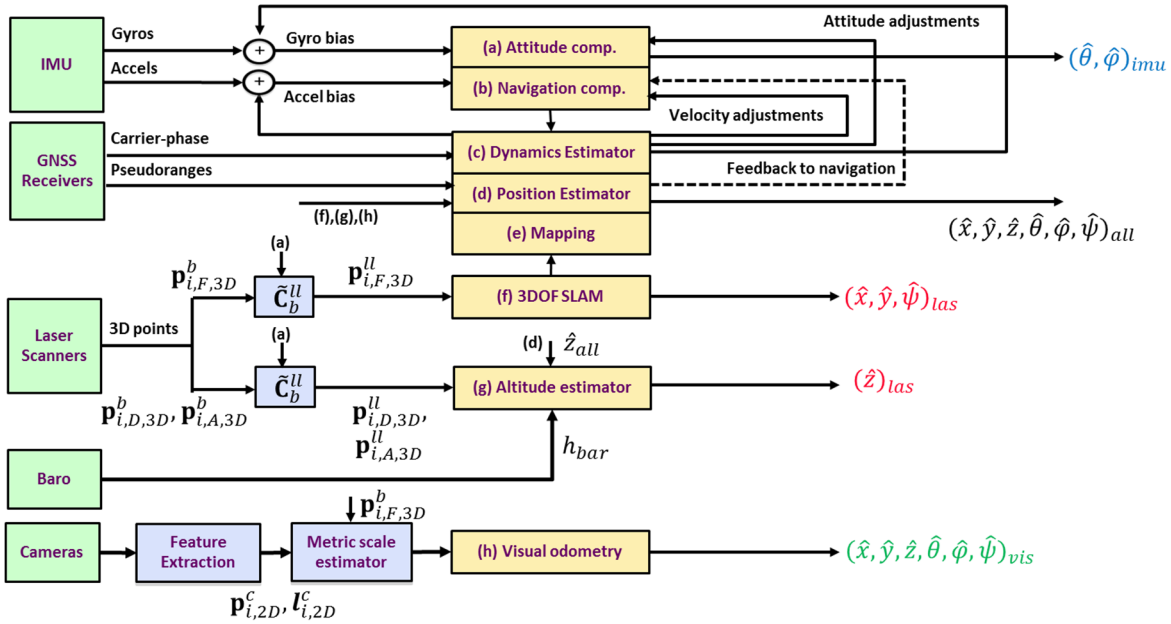


Figure 7.Functional block diagram of the proposed navigation and mapping method.

The dynamics estimator is a complementary Kalman filter (CKF) with error state vector:

$$\mathbf{x}_1 = [\delta\mathbf{v}^n \quad \delta\boldsymbol{\psi}_{nb} \quad \delta\boldsymbol{\omega}_{ib}^b \quad \delta\mathbf{f}_b] \qquad (1)$$

where $\delta\mathbf{v}^n$is the velocity error in the NED frame, $\delta\boldsymbol{\psi}_{nb}$is the miss-orientation vector, $\delta\boldsymbol{\omega}_{ib}^b$is the gyro bias vector, $\delta\mathbf{f}_b$is the specific force bias error. After the CKF update step, the dynamic filter error states are fed back to the attitude and navigation computation blocks. The measurement vector is given by the following error metric:

$$z_{,} = \mathbf{h}_j\Delta\mathbf{r}_{INS}^n - \Delta\varphi_j + a_j + b_j + \delta t_{rcvr} \qquad (2)$$

where $\Delta\varphi_j$ is the sequential carrier-phase difference for satellite 'j' $\Delta\mathbf{r}_{INS}^n$is the change in user position as computed by the inertial, $\mathbf{h}_j$ is the transpose of the line-of-sight vector to satellite 'j', ,$a_j$and $b_j$are two compensation terms for geometry and Doppler change correspondingly [14], and δt_{rcvr} is the sequential clock drift error. Note that equation (3) can be extended to a "difference-of-difference" measurement by taking the difference of the sequential difference of satellite 'j' and the sequential difference of a key satellite 'k'. This new

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.

difference term will remove the receiver clock drift error from the state vector.

The position estimator is a KF whose state vector is given by $\mathbf{x}_2 = \mathbf{r}_{INS}^n = [x \quad y \quad z]^T$ and has an extremely simple form that includes a forcing function from the dynamics estimator.

$$\mathbf{r}_{INS}^n(t_m) = \mathbf{r}_{INS}^n(t_{m-1}) + (vector\ forcing\ function) \tag{3}$$

In the absence of GNSS while under the canopy, the navigator the estimator can use the outputs of the laser scanners to compute the position. In our method, range and angle measurements (ρ_i, α_i) from the three laser scanners are converted three sets of 3D body-frame point cloud points by taking into account their lever arms and relative orientation with respect to the UAV reference point: $\mathbf{p}_{i,F}^b$, $\mathbf{p}_{i,A}^b$ and $\mathbf{p}_{i,D}^b$ Whereas the points from the angled and down-ward pointing laser scanned are used for terrain and altitude estimation, the points from laser F are used for computation of the 2D position following the flow shown in Figure 8.

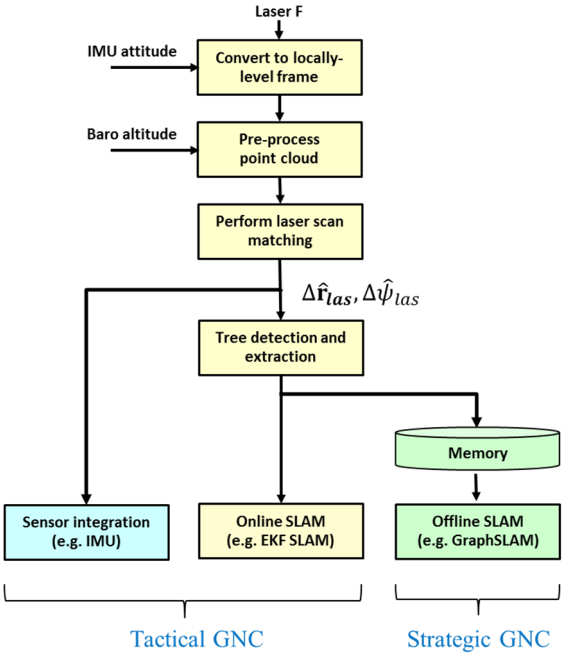


Figure 8. 3DOF Position estimator.

First, the points are converted to a locally-level frame using the attitude information from the module (a) in Figure 7. Before the actual laser scan matching stage, the point cloud points are pre-processed to remove sporadic returns, or points that are likely due to brush and ground returns. The remaining points are then input to a laser scan matching algorithm based on point-to-line iterative closest point (ICP) [22] or Generalized-ICP [23]. The result of the laser-scan matching algorithm in 2D is estimate of translation $\Delta\hat{\mathbf{r}}_{las}$ and incremental rotation $\Delta\hat{\psi}$. This estimated can be combined in the dynamics filter by defining a complementary error metric:

$$z_{,} = \Delta\mathbf{r}_{INS}^n - \Delta\mathbf{r}_{las}^n \tag{4}$$

Alternatively, tree features can be extracted from the processed points and input to an online SLAM algorithm like EKF SLAM [9]. In support of strategic GNC all point features and incremental poses (i.e. $\widehat{\Delta\mathbf{r}}_{LAS}, \Delta\widehat{\psi}_{LAS}$) can be stored in a graph like structure and optimized using a linear solver [21].

Figures 9, 10 and 11 show, respectively, input laser scan point cloud, pre-prosed point cloud and extracted tree features.

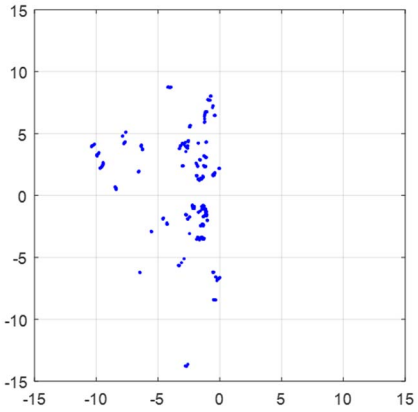


Figure 9. Input point cloud – includes clutter from environment

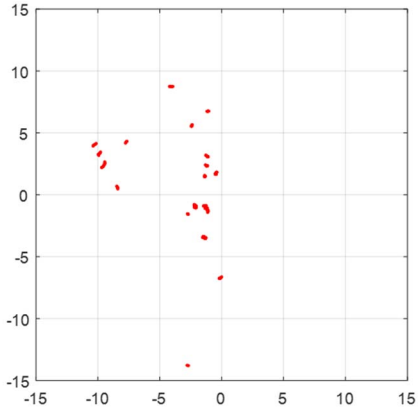


Figure 10. Processed point cloud – reduced clutter.

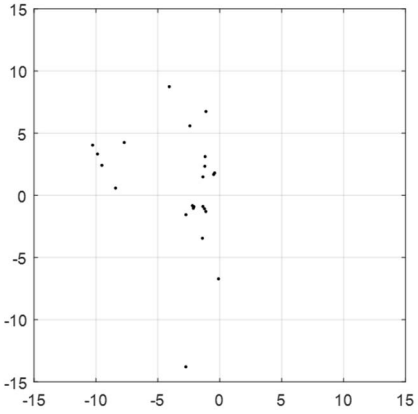


Figure 11. Tree feature extraction results.

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.

IV. ALTITUDE ESTIMATION

To obtain an estimate of the height, the downward-pointing and slanted laser scanners are utilized. Having multiple laser scanners rather than a mirror that deflects part of the forward-looking laser scanner, not only allows for a better estimate of the relative altitude, but also provides cross-track scans of the environment below the platform resulting in additional information for a 3D map. The terrain under the foliage typically has an irregular shape. Both laser scanners could be used in a similar fashion as was proposed in [24]: the angled laser scanner point cloud will be referenced to the navigation frame and used as the reference terrain database; the downward-pointing laser scanner will then observe part of the same terrain a couple of seconds later and generate its own navigation-frame referenced point cloud (see Figure 12). Matching of these two point clouds will result in an estimate of both the remaining horizontal position drift, a good estimate of height, and a rough terrain map. The extension of the method described in [24] to UAS operation in an under-the-foliage application will not be part of the discussion here.

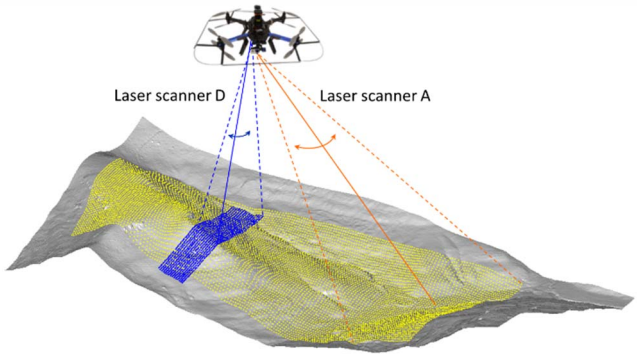


Figure 12. Altitude estimation using dual maps.

Instead, the height estimator assumes that the terrain under the foliage is partially structured and can be approximated by piecewise planar surfaces. The altitude estimator operates on data from both the downward and slanted laser scanners. Small subsets of these scans are taken from around their respective lowest points to form a planar surface that should be approximately orthogonal to a projected "plumb line" in the locally-level frame (see Figure 13). The normal point on this planar surface is assumed to lie at the intersection of the perceived floor and the platform's local-level z-axis, therefore, rendering the platforms relative altitude equal to the normal distance to the planar surface. To process the data further, possible outliers could be detected by applying more advanced detection schemes that make sure that the surface that it is being observed is truly a floor surface (i.e. flat lines in each scans subset of points). Alternatively, a simple Kalman filter could be designed to further reduce the noise and detect outliers by screening the filter residuals.

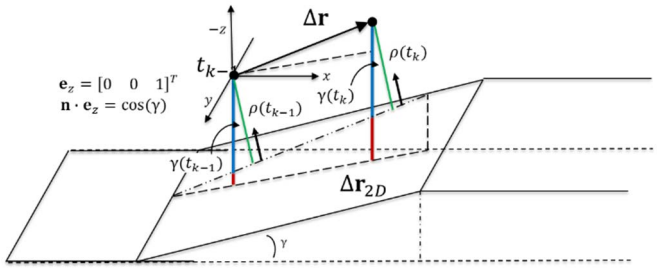


Figure 13. Altitude estimation using the planar assumption.

V. HARDWARE SETUP

The hardware platform and the location of the sensors with respect to the platform can be seen in Figure 14. The GPS receiver is a U-blox M8T L1 receiver capable of making both code-phase and carrier-phase measurements. The IMU is an Xsens MTI sensor which makes 3D acceleration, angular rate and magnetic heading measurements at an update rate of 100Hz. The camera consists of a Point Grey FireFly MV color camera running at 30 frames-per-second (fps).

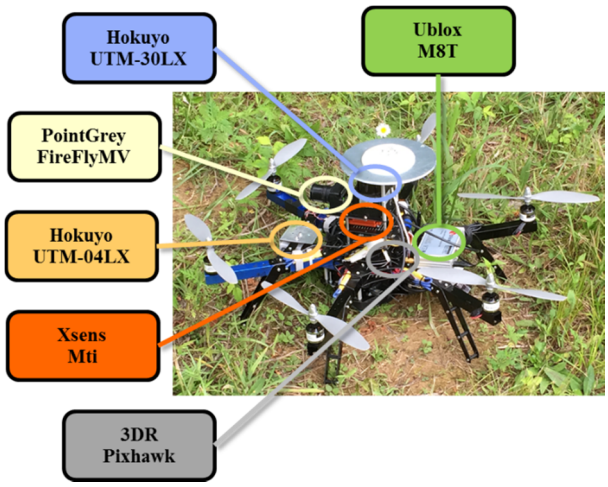


Figure 14. Hexacopter platform: sensor payload overview.

Finally, three laser scanners are mounted on the platform in the configuration shown in Figure 2. The forward-pointing laser scanner is a long-range (30m) Hokuyo UTM-30LX and the downward pointing laser scanners are both short-range (4m) Hokuyo URG-04LX units. The multi-copter is furthermore equipped with a PixHawk flight controller, which includes a baro-altimeter and gyros. Using either a RS232 or a USB for communication, each of the sensors is connected to a FitPC embedded processor, which is mounted just under the main sensor stack on the hexacopter. This FitPC contains an Intel Atom Z530 and runs the Robotic Operating System (ROS) on an Ubuntu operating system. For diagnostic purposes and communication with each of the sensors during flight, an 802.11b wireless connection is established between the FitPC

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.

and a ground station. During data collections, all of the desired sensor data is recorded on the FitPC, stored in "rosbags", and then offloaded to the ground station where ROS and Matlab are used to analyze the data.

VI. PRELIMINARY RESULTS

Two data collection missions were flown near Athens, Ohio, one on 15 April 2015 and one on 23 June 2015. Both missions were flown in a temperate forest with some relief in the terrain under the canopy. Both flights were performed under varying conditions as can be seen in Figure 15 and 16.



Figure15. Forest conditions for 15 April flights.



Figure16. Forest conditions for 23 June flights.

As the conditions for 23 June were more challenging, only data for that day is shown here. Figure 17 shows the GPS least squares position estimate from the Ublox M8T receiver. Initially, the multi-copter was stationary on the ground as indicated by (1) in the figure. The large errors here can be attributed to multipath, both specular reflections and diffraction from the trees. After take-off, the UAV enters the forest. Since the M8T is a high sensitivity receiver and the canopy is not completely closed, satellites are still observable, though at a much lower carrier-to-noise ratio resulting in higher noise levels as can be observed at various points along the track (2). Finally, the UAV leaves the forest and flies in open-sky conditions at (3). Lower noise is expected and observed here. Multipath effects are also reduced due to the UAV platform dynamics.

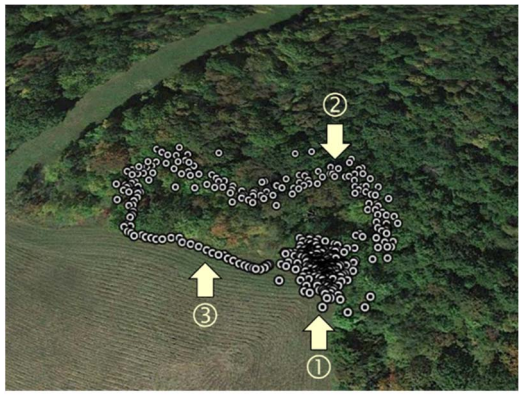


Figure 17. Ublox M8T trajectory.

The result of the pre-processing and laser-scan matching procedure described in Figure 8, are shown in Figure 18. A smooth trajectory can be observed due to the many features available in the relatively dense forest.

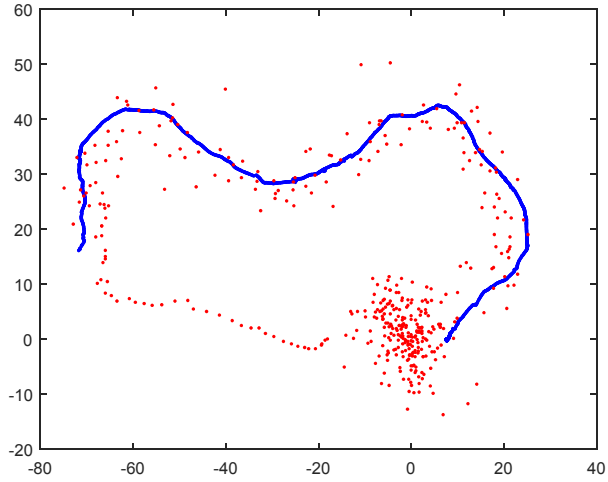


Figure18. Hexacopter platform: sensor payload overview.

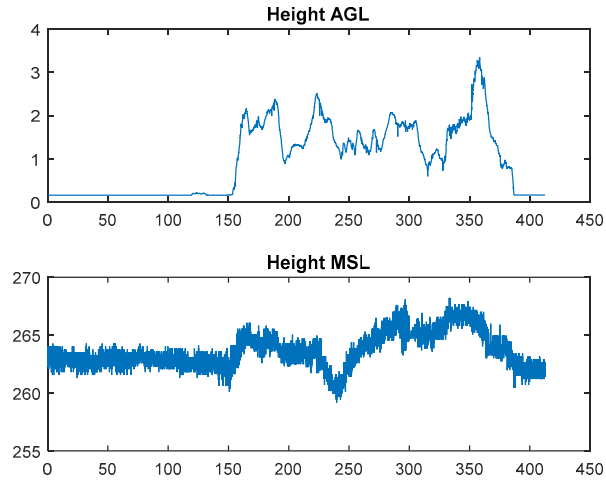


Figure19. Altitude profiles along the trajectory.

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.

Since, no truth reference was available, an absolute performance assessment could not be performed. However, the path seems consistent with the lower accuracy GPS path. Deviations are present in the path due to the large amount of clutter in the temperate forest environment as visible in Figure 15.

Figure 19 shows the altitude above ground level (AGL) and baro-altitude profile along the trajectory obtained from the method described in Section 4

VII. Summary and Conclusions

This paper describes a method for navigation in a forest environment. A multi-copter platform equipped with the aforementioned sensors has been designed, built and flight-tested in an appropriate temperate forest environment. Preliminary analysis of the collected data shows successful estimation of the UAV's 3D position and attitude with corresponding 2D tree maps. To get insight in the absolute performance of the algorithms, a truth reference environment will be setup in the form of a moderate forest mapped with a terrestrial laser scanner.

Current and future work include the development of an autonomous real-time flight capability; a terrain mapper based on the angled and down-ward pointing laser scanner as eluded to in Section 4; more advanced algorithms to reliably deal with the natural clutter (e.g. brush, small branches) in the forest environment; and a theoretical framework for the collision avoidance strategies.

Furthermore, this paper has omitted the use of the on-board vision camera, however, an integration method has already been developed and its evaluation using flight data will be part of a future paper.

Acknowledgment

This flight tests are funded in part by the Ohio University Russ Legacy Avionics Engineering Center (AEC) Development Fund. The team furthermore thanks Jonathan Sowash for letting us use various trails on his property for our flight test purposes.

References

[1] Farrell, J. L., *GNSS Aided Navigation & Tracking – Inertially Augmented or Autonomous*, American Literary Press, 2007.

[2] M. M. Miller, M. Uijt de Haag, A. Soloviev, M. Veth, "Navigating in Difficult Environments: Alternatives to GPS – 1," Proceedings of the NATO RTO Lecture Series on "Low Cost Navigation Sensors and integration Technology," SET-116, November 2008.

[3] M. M. Miller, J. Raquet, M. Uijt de Haag, "Navigating in Difficult Environments: Alternatives to GPS – 2," Proceedings of the NATO RTO Lecture Series on "Low Cost Navigation Sensors and Integration Technology," SET-116, November 2008.

[4] Nüchter, A., 3D Robotic Mapping – The Simultaneous Localization and Mapping Problem with Six Degrees of Freedom, Springer, 2010.

[5] Pfister, S. T., "Algorithms for Mobile Robot Localization and Mapping Incorporating Detailed Noise Modeling and Multi-scale Feature Extraction," Ph.D. Dissertation, California Institute of Technology, 2006.

[6] Soloviev, A., D. Bates, and F. van Graas, "Tight Coupling of Laser Scanner and Inertial Measurements for a Fully Autonomous Relative Navigation Solution," *NAVIGATION, Journal of the Institute of Navigation*, Vol. 54, No. 3, Fall 2007, pp. 189-205.

[7] Grisetti, G., C. Stachniss, W. Burgard, "Improving Grid-based SLAM with Rao-Blackwellized Particle Filters by Adaptive Proposals and Selective Resampling," *Proc. Of the IEEE Intl. Conf. on Robot. andAutom.*, Barcelona, Spain, April 2005, pp. 2432-2437.

[8] Rusinkiewicz, S. and M. Levoy, "Efficient variants of the ICP Algorithm," *Proc. Of Intl. Conf. on 3-D Digital Imaging and Modeling*, Quebec City, Quebec, May 2001, pp. 145-152.

[9] J. Guivant, et al., "Autonomous Navigation and Map building Using Laser Range Sensors in Outdoor Applications," *Journal of Robotics*, Vol. 17, No. 10, October 2000, pp. 565-583.

[10] Soloviev, A. and M. Uijt de Haag, "Three-Dimensional Navigation of Autonomous Vehicles Using Scanning Laser Radars: Concept and Initial Verification," *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 46, Issue 1, 2010.

[11] S. Kohlbrecher, O. von Stryk, J. Meyer, U. Klingauf, "A Flexible and Scalable SLAM System with Full 3D Motion Estimates," *Proc. Of IEEE Conference on Safety, Security, and Rescue Robotics*, 2011.

[12] S. Shen, N. Michael, V. Kumar, "Autonomous Multi-Floor Indoor Navigation with a Computationally Constrained MAV*," Proc. Of IEEE Conference on Robotics and Automation*, 2011.

[13] S. Grzonka, G. Grisetti, W. Burgard, "A Fully Autonomous Indoor Quadrotor," *IEEE Transactions on Robotics*, Vol. 28, Issue 1, 2012.

[14] E. Dill, "GPS/Optical/Inertial Integration for 3D Navigation and Mapping using Multi-copter Platforms," *Ph.D. Dissertation*, Ohio University, May 2015.

[15] Chrisholm, R. A., et al., "UAV LiDAR for below-canopy forest surveys," *Journal of Unmanned Vehicle Systems*, 2013.

[16] Cui, J. Q. et al., "Autonomous navigation of UAV in forest," International Conference on Unmanned Aircraft Systems, Orlando, FL, 2014

[17] Langelaan, J. and S. Rock, "Towards autonomous UAV flight in forests," Guidance, Navigation and Control Conference, San Francisco, 2005.

[18] Ross, S., et al., "Learning Monocular Reactive UAV Control in Cluttered Natural Environments," Robotics and Automation (ICRA), 2013 IEEE International Conference on, 1765-1772.

[19] A. Schultz, R. Gilabert, and M. Uijt de Haag, "Indoor Flight Demonstration Results of an Autonomous Multi-copter using Multiple Laser Inertial Navigation," Proceedings of the ION ITM 2016, Monterey, CA.

[20] Estrada, C., J. Neira, J. D. Tardo, "Hierarchical SLAM: real-time accurate mapping of large environments," *IEEE Transactions on Robotics*, Vol. 21, Issue 4, 2005.

[21] Kummerle, R., G. Grisetti, H. Strasdat, K. Konolige, W. Burgard, "g^2o: A general framework for graph optimization," Proc. of IEEE International Conference on Robotics and Automation (ICRA), 2011.

[22] A. Censi, "An ICP variant using a point-to-line metric" Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2008.

[23] A. Segal, D. Haehnel and S. Thrun , "Generalized-icp" , Robotics: Science and Systems V.

[24] Vadlamani, A. K., Airborne Laser Scanner Aided Inertial for Terrain Referenced Navigation in Unknown Environments, Ph.D. Dissertation, Ohio University, 2010.

Authorized licensed use limited to: University of Calgary. Downloaded on April 09,2026 at 06:41:42 UTC from IEEE Xplore. Restrictions apply.